

Khaled Lakhdher

AI / ML Engineer | Full-Stack AI Developer

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Sousse, July 23, 2026

[Hiring Manager / Recruiting Team]

[Company Name]

Subject: Application for the [Data Analyst / Data Specialist] position

Dear [Hiring Manager / Recruiting Team],

I am writing to express my strong interest in the [Data Analyst / Data Specialist] position at [Company Name]. As an AI / ML engineer focused on grounded LLM systems, retrieval-augmented generation, and full-stack delivery, I build AI that is reliable enough to ship, and I would be glad to bring that to your team.

For my final-year engineering project at CTT, I designed and built **Hotel Mind**, a conversational hotel-booking assistant on a grounded RAG architecture wired to a live SOAP reservation API. I engineered a hallucination-mitigation layer that verifies every factual claim so the assistant never invents a price or a booking, and delivered a containerized full-stack solution reaching nDCG@5 of 0.82 and MRR@5 of 0.90 on a benchmark query set. More recently I built **Nexus AI**, a multi-tenant platform that governs LLM agents like employees: scoped IAM-style permissions enforced at the tool-execution boundary, hierarchical vector memory, multi-agent orchestration over a Redis Streams message bus, and real-time monitoring, behind a provider-agnostic model gateway with per-agent cost tracking.

These projects reflect how I work: I take research-grade models and turn them into dependable products, with real attention to correctness, cost, and production concerns. Beyond deep learning and NLP, I am comfortable across the stack (Python, FastAPI, Django, React / Next.js, PostgreSQL / pgvector, and Docker), which lets me own a feature end to end, from model to deployment.

I am particularly drawn to [Company Name] because [what specifically draws you to this company: their product, mission, or the team you'd join]. I am confident that my experience building grounded, production-ready AI systems would let me contribute quickly and meaningfully to your goals.

I would welcome the opportunity to discuss how I can support [Company Name]'s work. Thank you for your time and consideration.

Sincerely,

Khaled Lakhdher